

EX:NO:32

PROGRAM:

```
count = 0
```

```
def merge_sort(arr):
```

```
    global count
```

```
    if len(arr) > 1:
```

```
        mid = len(arr)//2
```

```
        left = arr[:mid]
```

```
        right = arr[mid:]
```

```
        merge_sort(left)
```

```
        merge_sort(right)
```

```
    i=j=k=0
```

```
    while i < len(left) and j < len(right):
```

```
        count += 1
```

```
        if left[i] < right[j]:
```

```
            arr[k] = left[i]
```

```
            i+=1
```

```
        else:
```

```
            arr[k] = right[j]
```

```
            j+=1
```

```
        k+=1
```

```
    while i < len(left):
```

```
        arr[k] = left[i]
```

```
i+=1
```

```
k+=1
```

```
while j < len(right):
```

```
    arr[k] = right[j]
```

```
    j+=1
```

```
    k+=1
```

```
arr = [12,4,78,23,45,67,89,1]
```

```
merge_sort(arr)
```

```
print("Sorted:", arr)
```

```
print("Comparisons:", count)
```

Output

```
Sorted: [1, 4, 12, 23, 45, 67, 78, 89]
```

```
Comparisons: 16
```

```
=== Code Execution Successful ===
```